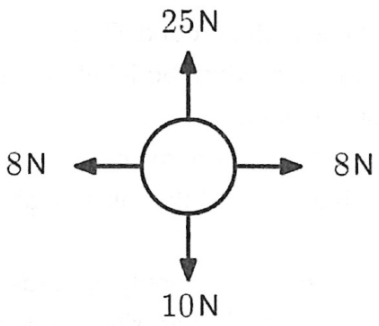
Forces

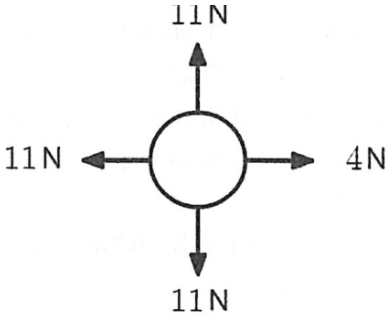
Questions 1–91

1.66 — Question 1. Find the resultant force in each of the following cases, stating both the value and direction:

a)

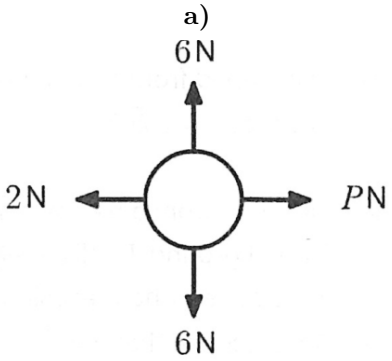


b)

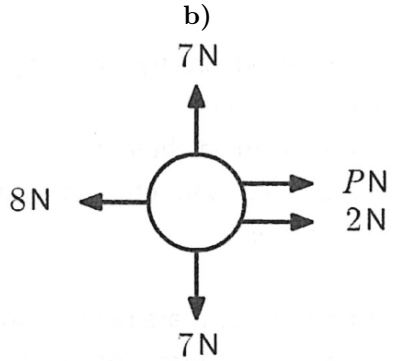


1.66 — Question 2. Given that each of these particles is stationary, find the value of P:

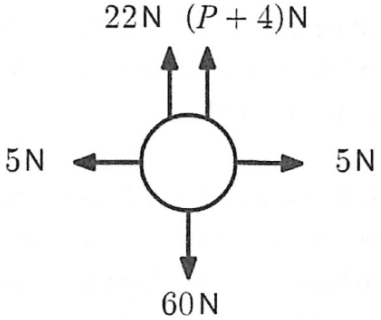
a)



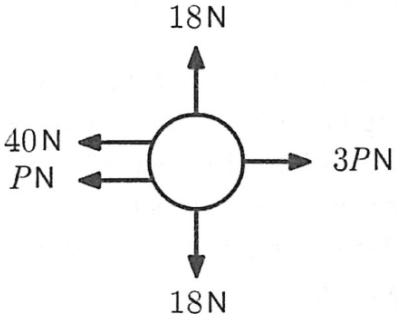
b)



c)

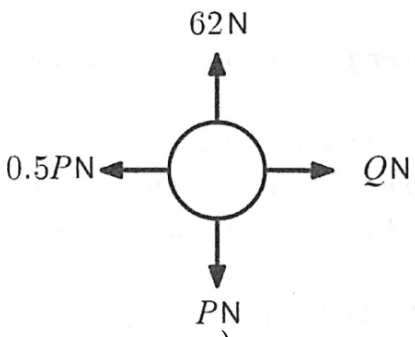


d)

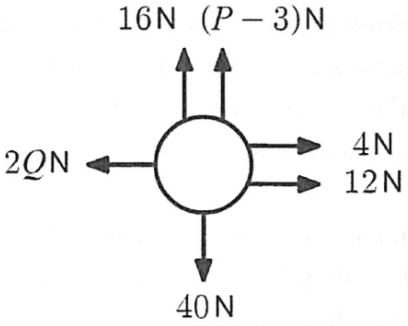


1.66 — Question 3. Find the value of P and Q such that the object is at rest.

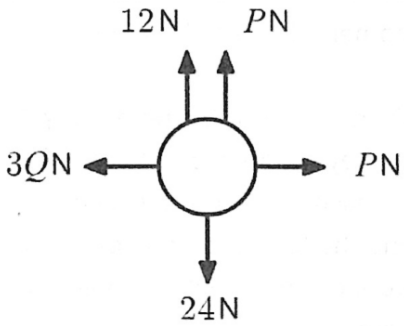
a)



b)

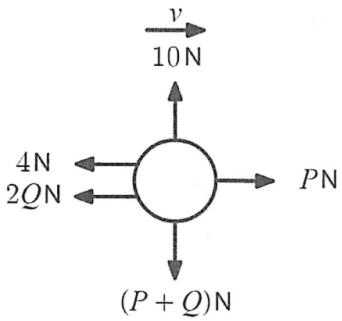


c)

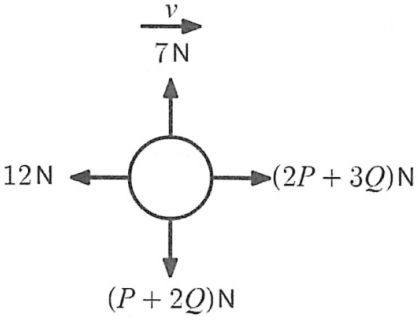


1.66 — Question 4. Each of the following diagrams shows an object acted on by a set of forces. Find the value of P and Q such that the object is moving with constant velocity.

a)



b)



1.66 — Question 5. An object of weight 50 N is hanging by a string in equilibrium as shown in the diagram.



a) State the value of T

b) By using $g = 10$ ms^{-2}, find the mass of the object.

The string snaps and the object drops.

c) State the size and direction of the resultant force acting on the object.

1.66 — Question 6. State the weight of an object of mass 50 kg

a) on earth given $g = 9.81$ ms^{-2}.

b) on the moon given $g = 1.63$ ms^{-2}.

1.66 — Question 7. An object has weight 4000 N on Earth where $g = 9.8$ ms^{-2}. Find the weight of the same object on the moon where the acceleration due to gravity is 1.6 ms^{-2}.

1.66 — Question 8. An object of mass 0.4 kg is being held in equilibrium by an upwards force of P N.
Using $g = 9.8$ ms^{-2}

a) Find the weight of the object.

b) Hence, state the value of P

The upwards force is removed.

c) State the size and direction of the resultant force acting on the object.

1.66 — Question 9. A small body is hanging in equilibrium at the end of a vertical string.

a) Draw a force diagram to represent the forces acting on the body.

Find, using $g = 9.8$ ms^{-2}, the

b) weight of the body if the mass is 2 kg.

c) mass of the body if its weight is 34.3 N.

d) tension in the string if the mass is 0.5 kg.

e) tension in the string if the weight of the body is 14.7 N.

1.66 — Question 10. A box is at rest on a horizontal table.

a) Draw a force diagram to represent the forces acting on the box.

Find, using $g = 10$ ms^{-2}, the

b) weight of the box if the box is 1.5 kg.

c) mass of the box if its weight is 60 N.

d) mass of the box if the normal reaction of the table on the box is 5 N.

e) normal reaction of the table on the box if the mass of the box is 1.2 kg.

f) normal reaction of the table on the box if the weight of the body is 25 N.

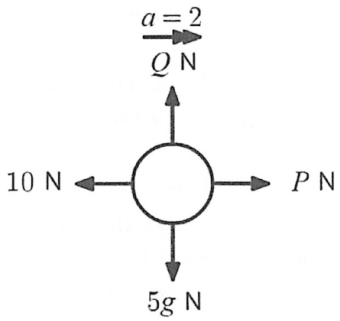
1.66 — Question 11. A book of mass 400 g is being pushed along with a force of 2 N on a smooth horizontal table.
Draw a force diagram to represent the forces acting on the book.

1.67 — Question 1. A particle mass m kg is acted upon by a resultant force F N has acceleration a ms^{-2}. In each case, find the unknown variable:

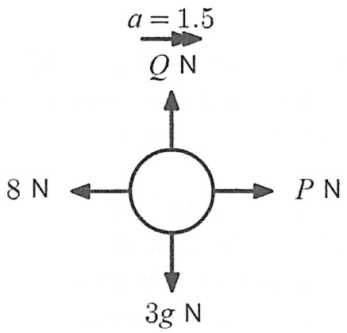
a) $m = 0.8$ kg, $a = 1.6$
b) $a = 0.8$, $F = 20$
c) $m = 5$ kg, $F = 15$
d) $m = 200$ g, $a = 0.8$

1.67 — Question 2. The following questions use $g = 9.8$ **ms**$^{-2}$ A particle of mass m kg with forces acting on it as shown in the diagram below, is accelerating at a ms^{-2}. In each case, state the value of m and find the values of P and Q

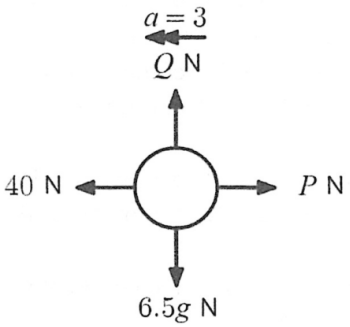
a)



b)

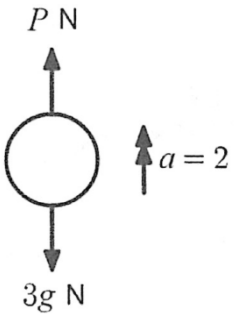


c)

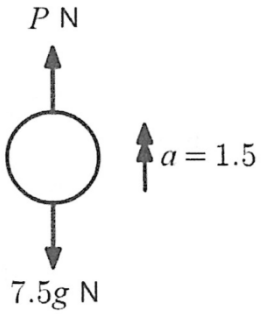


1.67 — Question 3. The following questions use $g = 10$ **ms**$^{-2}$ A particle of mass m kg with forces acting on it as shown in the diagram below, is accelerating at a ms^{-2}. In each case, state the value of m and find the value of P

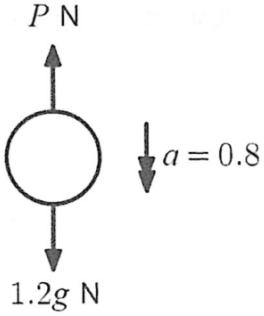
a)



b)



c)



1.67 — Question 4. A particle of mass 2 kg is attached to the end of a light, vertical string. The particle is accelerating vertically at a ms^{-2}. In each case find the tension in the string when:

a) $a = 3$ ms^{-2} upwards.
b) $a = 1.5$ ms^{-2} downwards.
c) $a = -2.5$ ms^{-2} upwards.
d) moving with constant velocity.

1.67 — Question 5. A particle of mass 0.5 kg is attached to the end of a light, vertical string. The particle is decelerating upwards at 0.2 ms^{-2}. Find the tension in the string.

1.67 — Question 6. The following questions use $g = 9.8$ **ms**$^{-2}$ A block of mass 8 kg is moving in a straight line with constant acceleration 0.1 ms^{-2}. The force driving it forward is 12 N. Find the total resistance acting against the motion of the block.

1.67 — Question 7. A box is being pushed on a horizontal surface such that it is moving with constant velocity. The pushing force is 100 N. Find the total resistive force.

1.67 — Question 8. A box is being pushed on a horizontal smooth surface by a 24 N force so that it is accelerating at 1.5 ms^{-2}.

a) Find the weight of the box.

b) State a modelling assumption used in your calculation.

1.67 — Question 9. A lorry of mass 3 tonnes is driving along a straight road. The lorry is experiencing resistance forces of 900 N. The engine is driving forward with a force of 2100 N. Find the acceleration of the lorry.

1.67 — Question 10. A pebble of mass 50 g is falling to solid ground through a still body of water. The pebble is experiencing a downwards acceleration of 2.5 ms^{-2}. Find the total resistive forces acting on the pebble.

1.67 — Question 11. A car of mass 2000 kg reduces its speed from 18 ms^{-1} to rest in 5 seconds, along a straight flat road.

a) Find the breaking force required to achieve this.

b) State the modelling assumption you have made in calculating the breaking force.

1.67 — Question 12. A vehicle of mass 900 kg reduces its speed from 20 ms^{-1} to rest in 6 seconds, along a straight flat road. It experiences a resistive force of 450 N during this time. Find the breaking force of the vehicle.

1.67 — Question 13. Prerequisite knowledge for Q13-Q18: Constant Acceleration formulae A stone of mass 400 g is dropped from a bridge and falls vertically. It hits the ground 1.5 seconds after being released. A constant resistance to motion of the stone is given to be 2 N. Find the height from which the stone is dropped.

1.67 — Question 14. A body of weight 20 N is accelerating along rough horizontal ground, being pushed by a force of 7 N acting parallel to the ground. A constant resistance force of 5.5 N acts against the particle.

a) Find the mass of the particle.

b) Given the particle starts at rest, find its speed after 10 seconds.

1.67 — Question 15. A block of mass 3 kg starts at rest and is accelerating across a smooth horizontal ground by a force of 1.5 N acting parallel to the ground. Find how far it has travelled in the first 4 seconds of motion.

1.67 — Question 16. A car of mass 1650 kg is driving with constant speed 10 ms^{-1} when it begins accelerating. After 30 m it reaches a speed of 18 ms^{-1}. Find the resultant force acting on the car during the period of acceleration.

1.67 — Question 17. A car of mass 2000 kg is driving at 47 km h^{-1} when the driver notices an accident ahead. When the car is 20 metres away from the accident, the driver applies the brakes creating a total resistance force of 10 000 N. Determine whether the car stops before the accident.

1.67 — Question 18. A van of mass 2250 kg drives a distance of 40 metres along a straight road, accelerating uniformly from rest to 25 ms^{-1}. Find

a) the acceleration of the van.
b) the magnitude of the force exerted by the engine, stating any modelling assumptions you have made in this calculation.
c) the magnitude of the normal reaction between the van and the road.

1.68 — Question 1. Two forces $\mathbf{P}$ and $\mathbf{Q}$ act on a particle. The vectors $\mathbf{i}$ and $\mathbf{j}$ are unit vectors due east and north respectively. Find the magnitude and direction (anticlockwise from the positive x-axis) of the resultant force in each case, giving your answers to 3 significant figures:

a) $\mathbf{P} = (2\mathbf{i} - \mathbf{j})$ N and $\mathbf{Q} = (-\mathbf{i} + 4\mathbf{j})$ N
b) $\mathbf{P} = \binom{3}{2}$ N and $\mathbf{Q} = \binom{5}{-4}$ N
c) $\mathbf{P} = (3\mathbf{i} + \mathbf{j})$ N and $\mathbf{Q} = (-5\mathbf{i} + 8\mathbf{j})$ N
d) $\mathbf{P} = \binom{-5}{-9}$ N and $\mathbf{Q} = 4\binom{1}{2}$ N
e) $\mathbf{P} = (-0.5\mathbf{i} + 2\mathbf{j})$ N and $\mathbf{Q} = 3(\mathbf{i} - 6\mathbf{j})$ N

1.68 — Question 2. Two forces **P** and **Q** act on a particle. The vectors **i** and **j** are unit vectors due east and north respectively. Find the magnitude and the bearing of the resultant force in each case:

a) $\mathbf{P} = (5\mathbf{i} + 3\mathbf{j})$ N and $\mathbf{Q} = (-\mathbf{j})$ N
b) $\mathbf{P} = \begin{pmatrix} -5 \\ 2 \end{pmatrix}$ N and $\mathbf{Q} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ N
c) $\mathbf{P} = (2\mathbf{i} - \mathbf{j})$ N and $\mathbf{Q} = (3\mathbf{i} - 11\mathbf{j})$ N
d) $\mathbf{P} = \begin{pmatrix} 5 \\ -1.2 \end{pmatrix}$ N and $\mathbf{Q} = \begin{pmatrix} -6 \\ 1 \end{pmatrix}$ N
e) $\mathbf{P} = (1.25\mathbf{i} - 0.75\mathbf{j})$ N and $\mathbf{Q} = (-2.25\mathbf{i} - 1.75\mathbf{j})$ N

1.68 — Question 3. The force $\begin{pmatrix} k \\ 5 \end{pmatrix}$ N where k is a positive constant, has a magnitude of $\sqrt{29}$ N. Find the value of k

1.68 — Question 4. The force $(2k\mathbf{i} - 2k\mathbf{j})$ N where k is a constant, has a magnitude of 12 N and a bearing of 315°. Find the value of k

1.68 — Question 5. Two forces $\mathbf{P}$ and $\mathbf{Q}$ act on a particle. The vectors $\mathbf{i}$ and $\mathbf{j}$ are unit vectors due east and north respectively. Find a third force $\mathbf{R}$ such that the three forces are in equilibrium:

a) $\mathbf{P} = (-2\mathbf{i} + \mathbf{j})$ N and $\mathbf{Q} = (-3\mathbf{i} + 11\mathbf{j})$ N
b) $\mathbf{P} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$ N and $\mathbf{Q} = \begin{pmatrix} -4 \\ \sqrt{3} \end{pmatrix}$ N
c) $\mathbf{P} = (\sqrt{2}\mathbf{i} - \mathbf{j})$ N and $\mathbf{Q} = (-3\mathbf{j})$ N
d) $\mathbf{P} = \begin{pmatrix} k \\ 1 \end{pmatrix}$ N and $\mathbf{Q} = \begin{pmatrix} -k \\ 1 \end{pmatrix}$ N where k is a constant.
e) $\mathbf{P} = (a\mathbf{i} + b\mathbf{j})$ N and $\mathbf{Q} = (-a\mathbf{i} + 2b\mathbf{j})$ N where a and b are constants.
f) $\mathbf{P} = \begin{pmatrix} 2a \\ 6a \end{pmatrix}$ N and $\mathbf{Q} = \begin{pmatrix} b \\ 6b \end{pmatrix}$ N where a and b are constants.

1.68 — Question 6. The forces $\begin{pmatrix} a \\ 3 \end{pmatrix}$ N, $\begin{pmatrix} 2b \\ -a \end{pmatrix}$ N and $\begin{pmatrix} -6b \\ b \end{pmatrix}$ N act on an object which is in equilibrium. Find the values of a and b

1.68 — Question 7. Three forces, $(6\mathbf{i} + 10\mathbf{j})$ N, $(-3.5\mathbf{i} - 12\mathbf{j})$ N and $(-2.5\mathbf{i})$ N act on a particle. It begins at rest at the origin. The vectors $\mathbf{i}$ and $\mathbf{j}$ are unit vectors due east and north respectively.

a) Find the resultant force in the form $(a\mathbf{i} + b\mathbf{j})$ N where a and b are constants to be found.
b) Find the magnitude and the bearing of the resultant force.
c) Describe the motion of the particle.
d) State the additional force that would be needed for the particle to instead be at rest.

1.68 — Question 8. Three forces, $(12\mathbf{i} + 11\mathbf{j})$ N, $(3\mathbf{i} - 15\mathbf{j})$ N and $(a\mathbf{i} + b\mathbf{j})$ N act on a particle. The vectors $\mathbf{i}$ and $\mathbf{j}$ are unit vectors due east and north respectively. State the values of a and b and b such that the particle is moving due East.

1.68 — Question 9. Two forces $\mathbf{P}$ and $\mathbf{Q}$ act on a particle. Find the angle between the two forces:

a) $\mathbf{P} = (3\mathbf{i} + 4\mathbf{j})$ N and $\mathbf{Q} = (5\mathbf{i} + 12\mathbf{j})$ N

b) $\mathbf{P} = \binom{2}{3}$ N and $\mathbf{Q} = \binom{5}{1}$ N

1.68 — Question 10. Two forces are given by $\begin{pmatrix}1\\-7\end{pmatrix}$ N and $\begin{pmatrix}k\\3k\end{pmatrix}$ N, where k is a positive constant. Given that the resultant of these two forces is parallel to $\begin{pmatrix}-4\\7\end{pmatrix}$, find the value of k

1.68 — Question 11. Three forces, $\begin{pmatrix}1\\2\end{pmatrix}$ N, $\begin{pmatrix}-3\\8\end{pmatrix}$ N and $\begin{pmatrix}a\\b\end{pmatrix}$ N act on a particle. The vectors **i** and **j** are unit vectors due east and north respectively.

a) State the values of a and b such that the particle is moving due North.

It is given that $a = 8$

b) Find the value of b such that the particle is moving due North-East.

1.68 — Question 12. Find the unit vector of each of the following vectors:

a) $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$
b) $3\mathbf{i} - \mathbf{j}$
c) $\begin{pmatrix} -1 \\ 4 \end{pmatrix}$
d) $-8\mathbf{i} - 6\mathbf{j}$

1.68 — Question 13. The force $\mathbf{F}$ is parallel to $\overrightarrow{AB}$. Given the coordinates of A and B and the magnitude of $\mathbf{F}$, find $\mathbf{F}$ in each case:

a) $A(0,0),\ B(-3,-3),\ |\mathbf{F}| = 5$ N
b) $A(0,-2),\ B(3,0),\ |\mathbf{F}| = 4$ N
c) $A(1,2),\ B(5,6),\ |\mathbf{F}| = 8$ N

1.68 — Question 14. Prerequisite knowledge for Q14-Q23: Constant Acceleration A resultant force of $(3\mathbf{i} + 4\mathbf{j})$ N is pushing against a particle of mass 0.5 kg. The particle is initially at rest.

a) Find the magnitude of the acceleration of the mass

b) Hence find the time it takes for the particle to move 80 metres.

1.68 — Question 15. A resultant force of $\binom{1}{2}$ N is pushing against a particle of mass 2 kg. The particle is initially at rest.

a) Find the magnitude of the acceleration of the mass.

b) Hence find its speed of motion after 4 seconds.

1.68 — Question 16. Two forces given by $(5\mathbf{i}+\mathbf{j})$ N and $(-2\mathbf{i}+3\mathbf{j})$ N are pushing against a particle of mass 4 kg. The particle is initially at rest.

a) Find the magnitude of the acceleration of the mass.

b) Hence find the time it takes for the particle to move 10 metres.

1.68 — Question 17. A force of $\binom{5}{11.6}$ N is pulling on an object of mass 2kg which is falling under the influence of gravity, where $g = 9.8$ ms^{-2}. Find the magnitude of the acceleration of the object.

1.68 — Question 18. An object of mass 2 kg is dropped. A force of $(4.5\mathbf{i} + 13.6\mathbf{j})$ N is pulling on the object as it falls under the influence of gravity, where $g = 9.8$ ms^{-2}.

a) Find the magnitude of the object's acceleration.

b) Another force of $(-4.5\mathbf{i} + 5.5\mathbf{j})$ N is applied to the object. Find the magnitude of the object's acceleration now.

1.68 — Question 19. A resultant force $\mathbf{F}$ N parallel to $4\mathbf{i} - 3\mathbf{j}$ is pulling a box of mass 4 kg. The particle is initially at rest. It takes 2 seconds to travel 5 metres. Find the vector form of $\mathbf{F}$

1.68 — Question 20. A resultant force **F** N parallel to $8\mathbf{i} + 4\mathbf{j}$ is pulling a box of mass 3 kg. Initially it is moving at 1 ms^{-1} but reaches 4 ms^{-1} in 5 seconds. Find the vector form of **F**

1.68 — Question 21. Two vectors $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ N and $\begin{pmatrix} p \\ q \end{pmatrix}$ N are acting on a box of mass 5 kg which is initially at rest. The box moves in the direction $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$

a) Show that $p - 2q + 7 = 0$

Given that $p = 5$

b) Find the distance the box has travelled in 5 seconds.

1.68 — Question 22. Two vectors $3\mathbf{i} - \mathbf{j}$ N and $p\mathbf{i} + q\mathbf{j}$ N are acting on a box of mass 3 kg which is initially at rest. The box moves in the direction $\mathbf{i} + 5\mathbf{j}$

a) Show that $5p - q + 16 = 0$

Given that $q = 6$

b) Find the speed of the box after 3 seconds.

1.68 — Question 23. Two vectors $\begin{pmatrix} -1 \\ 4 \end{pmatrix}$ N and $\begin{pmatrix} p \\ q \end{pmatrix}$ N are acting on a box of mass 2 kg which is initially at rest. The box moves in the direction $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$

a) Show that $4p - 3q - 16 = 0$

It begins at position A with a speed of 2 ms^{-1}. Given that $p = -2$

b) Find the speed of the box after 3 seconds.

1.69 — Question 1. By modelling each object as a particle and ignoring air resistance, draw a diagram showing all of the forces acting on:

a) A car of mass M kg towing a caravan of mass m kg on a rough surface.
b) A train engine of mass M kg pulling two trucks of mass m kg on a rough surface.
c) A helicopter of mass M kg with a person of mass m kg hanging directly below.

1.69 — Question 2. A car of mass 2000 kg is towing a caravan of mass 500 kg along a straight flat road. The driving force produced by the engine is 600 N. By modelling both the car and caravan as having no resistance to motion, find

a) the acceleration of the car.
b) the tension in the tow bar.

1.69 — Question 3. A car of mass 2200 kg is towing a trailer of mass 400 kg along a straight flat road. The engine of the car is creating a driving force of 1100 N. The resistance force acting on the car is 72 N and the resistance force acting on the trailer is 16 N. Find

a) the acceleration of the car and trailer.
b) the tension in the tow bar.

1.69 — Question 4. Two particles, P and Q which are connected by a light, inextensible string have mass 6 kg and 1.5 kg respectively. Particle P is pulled along by a horizontal force of 60 N. Particles P and Q have resistances of 12 N and 3 N acting against them respectively.

a) Find the acceleration of the particles.
b) Find the tension in the string.
c) Explain how the modelling assumptions that the string is light and inextensible have been used.

1.69 — Question 5. Two particles, P and Q which are connected by a light, inextensible rod have mass 10 kg and 2 kg respectively. Particle P is pushed along a rough horizontal surface by a horizontal force of 30 N. Particles P and Q both experience resistances which are proportional to their masses. Given that they are accelerating at 0.5 ms^{-2}, find the thrust in the rod.

1.69 — Question 6. Two particles, P and Q which are connected by a light, inextensible string have mass 10 kg and m kg respectively. Particle P is pulled along a smooth horizontal surface by a horizontal force of 120 N which creates an acceleration of 3 ms^{-2}. Find

a) the value of m
b) the tension in the string.

1.69 — Question 7. Two particles, P and Q which are connected by a light, inextensible string have mass $5m$ kg and $2m$ kg respectively. Particle P is pulled along by a horizontal force of 37.8 N. Particles P and Q have resistances of 6 N and 2.4 N acting against them respectively. Find

a) the acceleration in terms of m
b) the size of the tension in the string.
c) Explain how the modelling assumptions that the string is light and inextensible have been used.

1.69 — Question 8. A train engine of mass 10000 kg is towing 2 identical trucks of mass 3500 kg along a flat straight track which are experiencing the same resistive forces. The driving force produced by the engine is 25 000 N. They are moving with constant acceleration 1.2 ms^{-2}. The engine has a resistance force of 2400 N acting against it. Find

a) the total resistance acting against each of the trucks.
b) the tension in each of the couplings.

1.69 — Question 9. A car is towing a caravan. The driving force created by the car is 300 N. A constant resistance force of 180 N acts against the caravan. They move at constant velocity. Find the total resistance acting on the car.

1.69 — Question 10. A car of mass 1800 kg is towing a caravan of mass 800 kg along a straight flat road. They start at rest and then reach 5 ms^{-1} after having travelled 20 metres. The resistance force acting on the car is 90 N and the resistance force acting on the caravan is 40 N. Find

a) the acceleration of the car and caravan.
b) the driving force by the engine of the car.
c) the tension in the tow bar.

1.69 — Question 11. Two particles, P and Q which are connected by a light, inextensible string have mass 20 kg and 6 kg respectively. Particle P is pulled by a force of 145.6 N. Particles P and Q have resistances of 60 N and 18 N acting against them respectively. The system starts from rest. Find

a) the acceleration of P.
b) the distance P moves over the first 5 s.
c) the tension in the string between P and Q.

1.69 — Question 12. A train engine of mass 7000 kg is towing a carriage of mass 2000 kg along a straight flat track. The driving force produced by the engine is 12 600 N. The resistance forces on the train engine and carriage are $700k$ N and $200k$ N respectively.

a) Show that $a = \frac{7}{5} - \frac{k}{10}$
b) Find the tension in the coupling between the train engine and the carriage.

Given they start at rest and reach 3 ms^{-1} in 5 s,

c) find the resistance on the carriage.
d) find the resistance on the train engine.

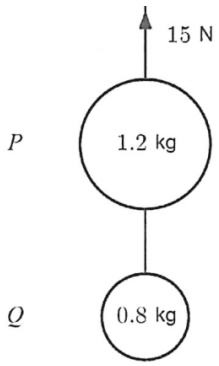
1.69 — Question 13. A van of mass 2500 kg is towing a trailer of mass 1000 kg along a straight flat road, with a tow bar. The driver reduces the speed of the van from 81 km h^{-1} to 63 km h^{-1} by using the van's brakes, creating a total force of 2590 N against the motion. The resistance force acting on the van is 150 N and the resistance force acting on the trailer is 60 N throughout the motion.

a) find the deceleration of the van.
b) find the time taken to slow to 63 km h^{-1}.
c) find the thrust in the tow bar.
d) State a modelling assumption you have made.

Whilst travelling at 63 km h^{-1} the tow bar breaks. The trailer slows to a rest without a collision. The total resistance to motion to the trailer is now 400 N.

e) find how far the trailer travels before coming to a stop.
f) Explain what the driver of the van would have noticed with the motion of the van.

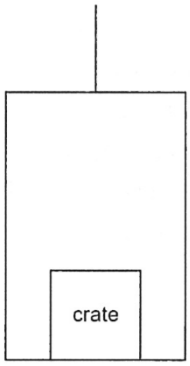
1.69 — Question 14. The following questions use $g = 9.8$ **ms**$^{-2}$ An object P of mass 1.2 kg is suspended by a light inextensible string. A second object Q of mass 0.8 kg is suspended by a second light inextensible string onto object P, as shown in the diagram. The first string has a force of 15 N applied, and the system moves upwards.



Find:

a) the acceleration of the system.
b) the tension in the lower string.
c) the value the 15 N needs to be increased by to achieve constant velocity.

1.69 — Question 15. For each of the following questions containing a crate in a lift, the following diagram may be used



A crate of mass m is in a lift which is travelling upwards. Find the magnitude of the force exerted on the crate by the lift when:

a) $m = 10$ kg and an acceleration 0.5 ms^{-2}
b) $m = 100$ g and an acceleration 0.7 ms^{-2}
c) $m = 2$ kg and a deceleration of 1.2 ms^{-2}
d) $m = 450$ g and a deceleration of 0.8 ms^{-2}

1.69 — Question 16. A lift of mass 800 kg is attached to a light, inextensible cable. It contains a crate of mass 100 kg.
Find the tension in the cable when the lift is:

a) Accelerating downwards at 0.5 ms^{-2}
b) Accelerating upwards at 0.5 ms^{-2}
c) Decelerating downwards at 0.4 ms^{-2}
d) Decelerating upwards at 0.4 ms^{-2}

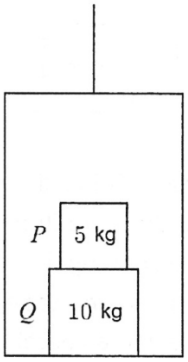
Find the acceleration upwards of the lift when the tension in the cable is:

e) 2700 N
f) 8100 N
g) 9450 N

1.69 — Question 17. The following questions use $g = 10$ **ms**$^{-2}$ A lift of mass 600 kg is attached to a light, inextensible cable. It contains a crate of mass 50 kg. The lift starts from rest and moves upwards, reaching the required floor which is 10 metres above in 15 seconds. Find

a) the acceleration of the lift.
b) the tension in the cable.
c) the magnitude of the force exerted on the crate by the floor of the lift.

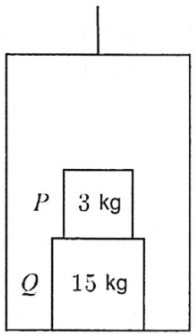
1.69 — Question 18. A lift of mass 100 kg accelerates upwards at 0.4 ms^{-2}, being pulled by a light inextensible vertical cable. The lift contains two boxes, P and Q of masses 5 kg and 10 kg respectively. Box P sits directly on top of box Q as shown in the diagram below:



Find

a) the tension in the cable.
b) the force exerted on Q by P
c) the force exerted by Q on the lift.

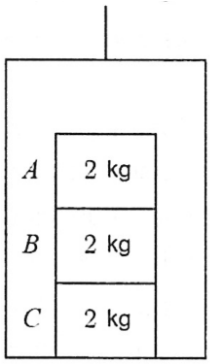
1.69 — Question 19. A lift of mass 120 kg decelerates upwards at 0.25 ms^{-2}, being pulled by a light inextensible vertical cable. The lift contains two crates, P and Q of masses 3 kg and 15 kg respectively. Crate P sits directly on top of crate Q



Find

a) the tension in the cable.
b) the force exerted on Q by P.
c) the force exerted by Q on the lift.

1.69 — Question 20. A lift of mass 120 kg accelerates upwards at 0.1 ms^{-2} pulled by a light inextensible vertical cable, carrying three identical crates, A, B and C, each with a mass of 2 kg.



Find

a) the tension in the cable.
b) the force exerted on the lift by C.

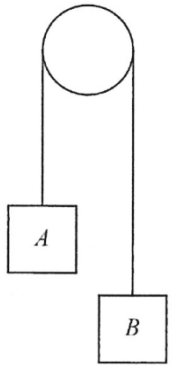
1.69 — Question 21. Prerequisite knowledge for Q21: Variable Acceleration Two particles P and Q which are connected by a light, inextensible string have mass 8 kg and 5 kg respectively. Particle P is being pulled along a smooth horizontal surface with velocity v ms^{-1} at time t seconds given by

$$v = -\frac{1}{5}t^2 + 6t, \quad 0 \leq t \leq 30$$

Find, at 10 seconds into the motion:

a) the acceleration of the particles.
b) the tension in the string.
c) the force pulling P.

1.70 — Question 1. The following questions use $g = 9.8$ ms^{-2} Two particles A and B of mass m_A and m_B are attached to the end of a light inextensible string which passes over a small smooth fixed pulley. The masses hang with the string taut, and the system is released from rest.



The system is released from rest. In each case find the tension in the string and the acceleration of the system:

a) $m_A = 9$ kg, $m_B = 3$ kg
b) $m_A = 2$ kg, $m_B = 8$ kg
c) $m_A = 200$ g, $m_B = 500$ g

1.70 — Question 2. Two particles A and B of masses $5m$ kg and $3m$ kg respectively are attached to the end of a light inextensible string which passes over a small smooth fixed pulley. The masses hang with the string taut, and the system is released from rest. Find

a) the acceleration of the system.
b) the tension in the string in terms of m
c) the magnitude of the force exerted on the pulley while A and B are in motion.

1.70 — Question 3. The following questions use exact values of g Two particles A and B of masses m kg and M kg respectively, where $M > m$, are attached to the end of a light inextensible string which passes over a small smooth fixed pulley. The masses hang with the string taut, and the system is released from rest. The tension in the string is T N and the acceleration is a ms^{-2}. Show that

$$a = \left(\frac{M-m}{M+m}\right) g$$

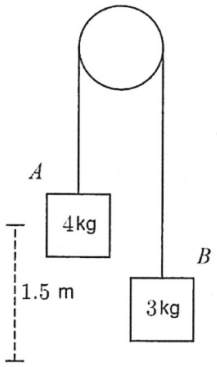
1.70 — Question 4. Two particles A and B of masses km kg and m kg respectively where $k > 1$. They are attached to the end of a light inextensible string which passes over a small smooth fixed pulley. The masses hang with the string taut, and the system is released from rest. The acceleration of the system is a ms^{-2}. Show that

$$a = \left(\frac{k-1}{k+1}\right) g$$

1.70 — Question 5. The following questions use $g = 9.81$ **ms**$^{-2}$ Two particles A and B are attached to the end of a light inextensible string which passes over a small smooth fixed pulley. The masses hang with the string taut, and the system is released from rest. A has mass 2 kg and descends 0.8 metres in the first 4 seconds after the system is released. Find

a) the acceleration of the system.
b) the tension in the string.
c) the mass of B.

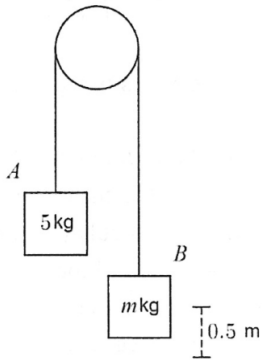
1.70 — Question 6. Two particles A and B of masses 4 kg and 3 kg respectively are attached to the end of a light inextensible string which passes over a small smooth fixed pulley.



The masses hang with the string taut, and the system is released from rest. Initially, particle A begins 1.5 metres above the ground.
Find

a) the acceleration of the system.
b) the tension in the string.
c) the time it takes for A to hit the ground.
d) the greatest height of B above the floor, stating any modelling assumptions you have made.

1.70 — Question 7. The following questions use $g = 9.8$ ms^{-2}
Two boxes A and B of masses 5 kg and m kg respectively are attached to the end of a light inextensible string which passes over a smooth small fixed pulley. The masses hang with the string taut, and the system is released from rest.



Box B starts 0.5 metres above the ground.
Box A ascends 0.2 metres in the first 0.5 seconds after the system is released.

a) Find the acceleration of the system.
b) Find the tension in the string.
c) Show that the mass of B is approximately 7 kg.
d) Find the velocity at which B hits the ground.

1.70 — Question 7 (continued). Once Box B hits the ground, the string holding up box A becomes slack.

e) Given box A starts 1 metre above the ground, find the greatest height that box A reaches above the ground.

f) Find the time box A spends with the string not taut.

A different box (box C) is attached to the string, in replacement of box B. The system is re-set so that Box C starts 0.5 m above the ground.

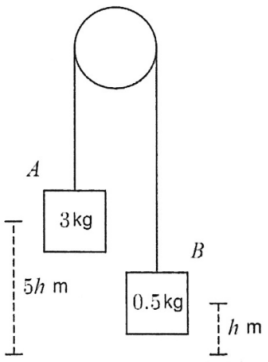
Explain what box A will experience with its acceleration (including the direction) and the tension in the string in each of the following scenarios:

g) Box C has a larger mass than box B.

h) Box C has the same mass as box A.

i) Box C has a lower mass than box A.

1.70 — Question 8. Two boxes A and B of masses 3 kg and 0.5 kg respectively are attached to the end of a light inextensible string which passes over a small smooth fixed pulley. The system is released from rest. Box A begins $5h$ metres from the ground and box B begins h metres from the ground.



Find

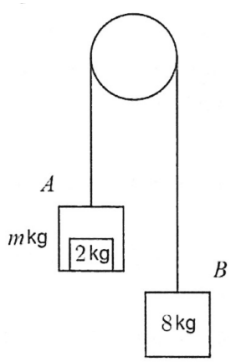
a) the acceleration of the system.

b) the tension in the string.

c) Show that the time after release that both boxes are at the same height is given by

$$t = \sqrt{\frac{4h}{7}}$$

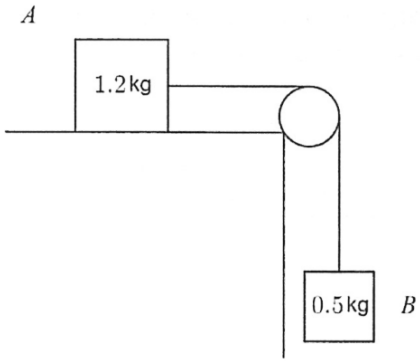
1.70 — Question 9. A scale pan A of mass m kg hangs freely and is attached by a light inextensible string to a mass B which is 8 kg. The string passes over a small smooth fixed pulley. The masses hang with the string taut.



An object of mass 2 kg is placed into the scale pan and the system is released from rest. The scale pan A descends but doesn't exceed 1.96 ms^{-2}. Find

a) the range of the values of T where T N is the tension in the string.
b) the range of values of m

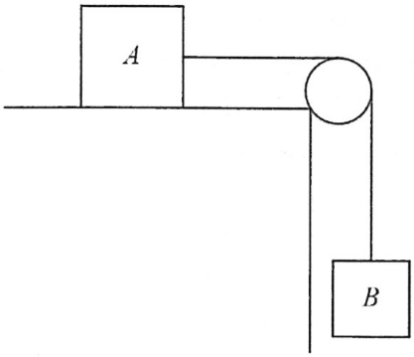
1.70 — Question 10. Two particles, A and B have masses 1.2 kg and 0.5 kg respectively. They are connected by a light, inextensible string. Particle A lies on a smooth horizontal table and the string passes over a small round smooth pulley which is fixed at the edge of a table. Particle B hangs freely.



Find

a) the acceleration of the system.
b) the tension in the string.

1.70 — Question 11. Two particles, A and B have masses m_A and m_B respectively. They are connected by a light, inextensible string. Particle A lies on a rough horizontal table and the string passes over a small round pulley which is fixed at the edge of a table. Particle B hangs freely. The resistance force acting on particle A is F N.



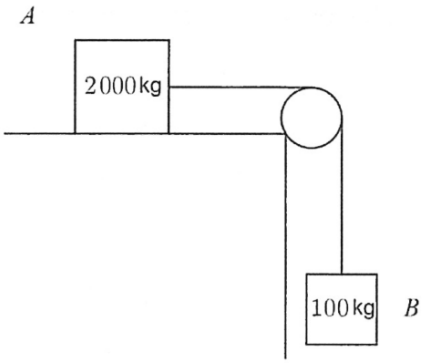
The system is released from rest.
Find, in each case, the acceleration of the system and the tension in the string.

a) $m_A = 2$ kg, $m_B = 3$ kg, $F = 20$
b) $m_A = 5$ kg, $m_B = 7$ kg, $F = 12$
c) $m_A = 200$ g, $m_B = 500$ g, $F = 2$
d) $m_A = 8$ kg, $m_B = 6$ kg, $F = 58.8$

1.70 — Question 12. For each of the scenarios in question 11), find the time it takes for particle B to travel 1 m, assuming particle B begins over 1 metre from the floor.

1.70 — Question 13. The following questions use $g = 10$ ms^{-2}

A van A of mass 2000 kg is attached with a cable to a pallet of building materials B which has mass 100 kg. The cable passes over a smooth fixed pulley, and B hangs freely.



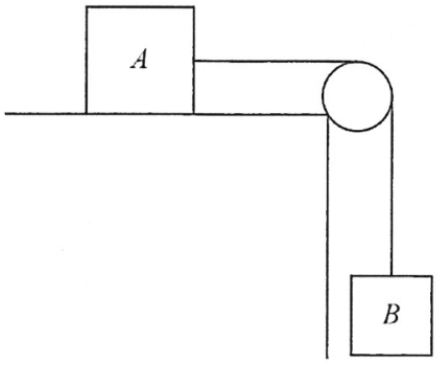
Initially the system is at rest and the cable is taut. The van begins to drive away from the pulley with driving force 2500 N. The resistance force acting on the van is 450 N.

a) Find the acceleration of the system.
b) Find the tension in the cable.
c) State two modelling assumptions that you have made.

The van drives forward for 10 seconds before the pallet is getting close to reaching the pulley.

d) Find the height that the pallet has been lifted.
e) State any assumptions in you have used.

1.70 — Question 14. Two particles, A and B have masses M kg and m kg respectively. They are connected by a light, inextensible string. Particle A lies on a smooth horizontal table and the string passes over a small round smooth pulley which is fixed at the edge of a table. Particle B hangs freely.

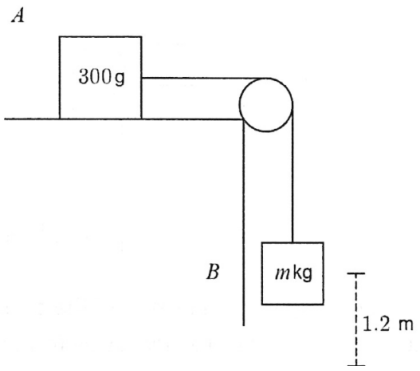


Show that

$$a = \frac{10m}{m + M}$$

1.70 — Question 15. The following questions use $g = 9.8$ ms^{-2}

A wooden block A with mass 300 g sits on a rough table, attached to a string which passes over a small, smooth pulley which is fixed to the edge of the table. A plastic block B of mass m kg hangs freely at the end of the string.



The system is released from rest and block B descends. The resistance to motion on block A is 5 N.

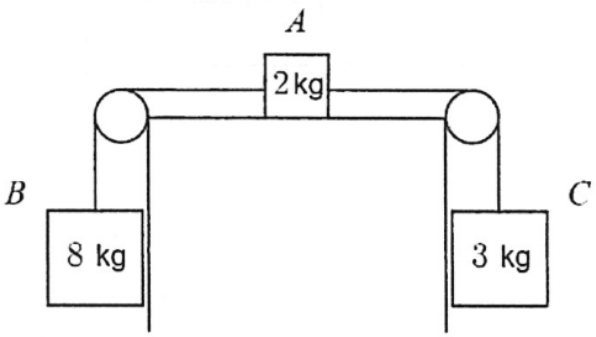
a) Find the acceleration of the system in terms of m

The block B initially starts 1.2 metres above the ground. It takes 2 seconds for B to reach the ground.

b) Find the value of m

c) State any modelling assumptions you have made.

1.70 — Question 16. Particles A, B and C, have masses 2 kg, 8 kg and 3 kg respectively. A and B are attached to the ends of a light inextensible string. B and C are attached to the ends of another light inextensible string. B rests on a smooth table. A and C hang freely. Both strings pass over small smooth fixed pulleys at the edge of the table with the string taut.



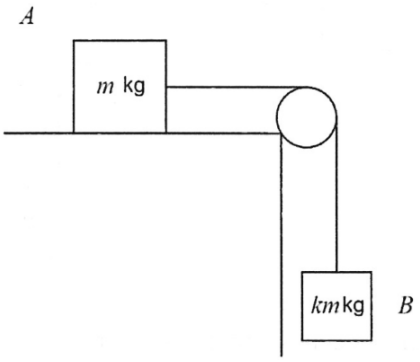
The system is released from rest. Find

a) the acceleration of the system.

b) the tension in the two strings.

1.70 — Question 17. The following questions use exact values of g

Two particles, A and B have masses m kg and km kg respectively where $k > 1$. They are connected by a light, inextensible string. Particle A lies on a rough horizontal table and the string passes over a small round smooth pulley which is fixed at the edge of a table. Particle B hangs freely.

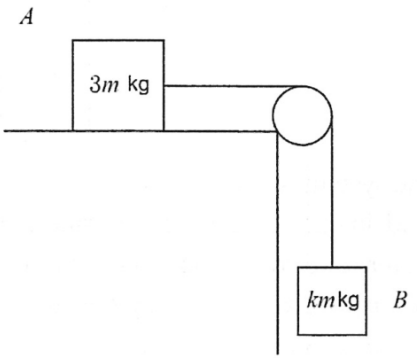


The system is released from rest.

The resistance to motion on particle A is given as $\frac{1}{2}kmg$ N. Show that

$$a = \frac{kg}{2(1+k)}$$

1.70 — Question 18. Two particles, A and B have masses $3m$ kg and km kg respectively where $k > 0$. They are connected by a light, inextensible string. Particle A lies on a rough horizontal table and the string passes over a small round smooth pulley which is fixed at the edge of a table. Particle B hangs freely.



A horizontal force of $20mg$ N is applied to A away from the pulley. The resistance to motion on particle A is given as $10mg$ N. Initially at rest, particle A travels 0.5 m in 10 s.

a) Find the acceleration of the system

b) Show that

$$k = \frac{1000g - 3}{100g + 1}$$